# 1 OPTIMISE WORK PRODUCTIVITY WITH CLEAN INDOOR AIR

Even the cleanest looking environment may have invisible particles that can deteriorate health and aggravate health issues. Clean air is defined as air that has no harmful pollutants, such as dust, dirt, airborne microbes or chemicals, in it. This is why we require air purifiers to reduce the pollutants in the air around us.

We exhale carbon-dioxide, causing indoor carbon-dioxide and other air pollutants to increase. Besides being exposed to distracting, irritating and potentially unhealthy gases and particulates, elevated carbon-dioxide levels can infuse lethargy and drowsiness among us.

Long-term exposure to high levels of carbon-dioxide could also induce a condition where people in a building feel sick for no apparent reason. Poor ventilation, fumes, fabric fibres, dust particles, bright/flickering lights or a combination of all these factors are likely causes of this condition.

Air Ok Technologies, an IIT-Madras-incubated clean-tech startup, has launched Vistar comfort air purifiers with efficient granular adsorbent particulate arrester (EGAPA) technology to promote a healthy work environment for improved productivity. The purifiers are built with special media to absorb carbon-dioxide and other acidic gases in the air to make the workplace atmosphere cleaner, healthier and livelier. These can effectively optimise air quality by reducing carbon-dioxide levels for a coverage area of 15m$^2$ to 104m$^2$.

Vishesh Kaul, head sales and operations, Air Ok Technologies, states, "Increased levels of carbon-dioxide adversely impact people, including their thinking capabilities, energy levels and more. Besides providing a comforting air-conditioned environment, it is also necessary for companies to improve quality of indoor air that allows employees to be more active and participative in business activities."

EGAPA filters help in maintenance of better air quality by removing up to 99.7 per cent harmful indoor air pollutants. These have high-performance filtration rates, which help them adsorb all acidic and alkaline gases. Air enters the purifiers through slits angled at 45 degrees and moves to obtain a high mass transfer zone. The purifiers have the capability of filtering out particles up to 0.3 microns along with other such basic pollutants as formaldehyde, allergens, moulds and airborne bacteria.

The air purifiers find applications for human and hardware health in data centres and server rooms. Protecting datacom equipment and servers from any potential contaminants threat is a vital step in ensuring their good health and continued viability.

The devices are easy-to-operate with uni-touch interface, where control is at fingertips. Their low-noise generation, Wi-Fi and cloud connectivity make them a smart choice for air purification.

The user can also control the functions via a mobile app for real-time pollution monitoring. It demands low maintenance with filter replacement every eight to twelve months depending on carbon-dioxide concentration levels.

The air purifiers are priced at ₹ 25,000 to ₹ 200,000, based on their features and capabilities. The devices ensure that the air you breath is safe and healthy with a customised solution for every air quality need, residential as well as commercial.

*—Deepshikha Shukla*